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May 1st, 2026

Molecular Signatures of Breast Cancer: A TCGA-BRCA Differential Expression Analysis

Introduction

Breast cancer is the most commonly diagnosed among women worldwide, affecting approximately 1 in 8 women over their lifetime (American Cancer Society, 2024). Despite significant advances in early detection and targeted therapies, breast cancer remains a leading cause of cancer-related mortality, in large part due to its biological complexity. Breast tumors are particularly well-characterized for their molecular heterogeneity. They are formally classified into distinct subtypes (Basal, Her2-enriched, Luminal A, and Luminal B) based on gene expression patterns, each carrying different clinical prognoses and treatment implications. Even within a single subtype, individual tumors can exhibit substantially different gene activity profiles. This heterogeneity poses a fundamental challenge for diagnosis and treatment, as therapeutic strategies that are effective for one patient may be ineffective for another.

One of the most powerful approaches for characterizing this complexity at the molecular level is transcriptomic profiling through RNA sequencing (RNA-seq), which measures the activity level of every gene across a tissue sample simultaneously (Wang et al., 2009). By comparing gene expression between malignant and healthy tissue, researchers can identify differentially expressed genes (DEGs), which are genes that are significantly turned on or off in cancer cells relative to normal cells. These DEGs serve as candidate biomarkers, or molecular signals that may reflect the underlying biology of tumor development and could inform more precise diagnostic or therapeutic strategies.

The goal of this analysis was to identify which genes are most significantly dysregulated between primary tumor tissue and healthy adjacent tissue in breast cancer, as well as exploring whether expression of key candidate genes varies across established molecular subtypes. To address this, RNA-seq data from The Cancer Genome Atlas (TCGA) Breast Invasive Carcinoma (BRCA) cohort was analyzed using a differential expression pipeline implemented entirely in R Studio, leveraging Bioconductor packages that are built for high-throughput genomic data. This dataset represents one of the largest and most well-characterized breast cancer cohorts publicly available, making it a robust foundation for transcriptomic discovery.

Data & Methods

1. Data Acquisition

RNA-seq data from the TCGA-BRCA cohort was retrieved directly into R Studio using the TCGAbiolinks Bioconductor package (Colaprico et al., 2016), which provides a programmatic interface to the NIH Genomic Data Commons (GDC) portal. The query was restricted to two sample types: Primary Solid Tumor (TCGA sample code 01) and Solid Tissue Normal (TCGA sample code 11), yielding 1,111 tumor samples and 113 normal samples. The data type retrieved was STAR-generated raw gene expression counts, where reads from each sample are mapped to the human reference genome and summarized as integer counts per gene. The resulting dataset, prepared as a SummarizedExperiment object using GDCprepare(), contained expression measurements for 60,660 genes across 1,224 samples,

along with clinical metadata including PAM50 molecular subtype, AJCC pathological stage, and vital status.

The imbalance between tumor ($n = 1,111$) and normal ($n = 113$) samples is a known structural limitation of the TCGA-BRCA cohort, not an artifact of the query. Collecting matched normal tissue requires an additional biopsy from healthy adjacent tissue at the time of surgery, which is not always clinically performed. Importantly, the 113 normal samples are not guaranteed to be matched pairs with a corresponding tumor sample from the same patient. The final model therefore treated tumor and normal as independent groups rather than explicitly paired observations, which is a limitation acknowledged in the context of the original proposal's paired design goal.

2. Pre-processing and Filtering

Prior to modeling, low-count genes were removed using a minimum count filter. Any gene with a total read count summed across 1,224 samples below 10 was excluded. This step is standard practice in RNA-seq analysis and serves two purposes: it removes genes that lack sufficient evidence for reliable statistical testing, and it reduces the multiple testing burden, improving overall statistical power. This reduced the dataset from 60,660 to 56,709 genes carried forward into differential expression modeling.

Differential expression analysis was performed using DESeq2 (Love et al., 2014), a Bioconductor package built for count-based RNA-seq data. DESeq2 was selected over alternatives such as edgeR or limma-voom for several reasons. First, it uses a negative binomial model to account for overdispersion, a property of RNA-seq count data where variance exceeds the mean, which violates the assumptions of simpler models like Poisson regression. Second, DESeq2 applies its own internal normalization using median-of-ratios, which corrects for differences in sequencing depth across samples without requiring a separate normalization step. Third, DESeq2 uses shrinkage estimation for dispersion and fold change, which stabilizes results particularly for genes with low counts or high variability.

The model was specified as $\sim$ sample_type, comparing Primary Solid Tumor against Solid Tissue Normal as the reference level. DEGs were identified using a dual threshold: a Benjamini-Hochberg adjusted p-value below 0.01 and an absolute log2 fold change greater than 2. The adjusted p-value threshold controls the false discovery rate (FDR) rather than the raw type I error rate, which is important given the scale of simultaneous testing across tens of thousands of genes. The log2fold change threshold of 2, equivalent to a 4-fold difference in expression, ensures that identified genes are not only statistically significant but biologically meaningful in magnitude.

3. Transformation and Visualization

For downstream visualization, raw counts were transformed using DESeq2's Variance Stabilizing Transformation (VST), which addresses a key property of count data: raw counts are heteroscedastic, meaning that genes with higher expression tend to show higher variance simply due to scale. VST normalizes this relationship so that variance is approximately independent of the mean, making the transformed values appropriate for distance-based methods such as PCA and hierarchical clustering. Ensembl gene IDs were mapped to gene symbols using the org.Hs.eg.db annotation database to improve interpretability of figures.

Four visualizations were produced. A PCA plot and volcano plot were generated using ggplot2 and EnhancedVolcano respectively, a clustered heatmap of the top 50 DEGs was produced using ComplexHeatmap with the viridis color palette, and stratified boxplots of MMP11 expression by PAM50 subtype were generated using ggplot2. For the heatmap, the full cohort was downsampled to 100 tumor

and 100 normal samples ($n = 200$) using a fixed random seed (`set.seed(42)`) to preserve reproducibility while improving visual clarity.

Results

1. PCA Confirms Tumor vs. Normal Separation

Principle Component Analysis was performed on VST-normalized expression data across all 1,224 samples to assess whether global gene expression patterns were sufficient to distinguish tumor from normal tissue before proceeding with the formal statistical testing. As shown in Figure 1, PC1 (19% variance) and PC2 (12% variance) together account for 31% of total transcriptomic variance, and samples separate broadly along PC1 by sample type. Normal samples (blue, $n = 113$) form a tight, well-defined cluster in the upper-left quadrant, indicating low variability and biologically consistent expression profiles across healthy tissue. Tumor samples (red, $n = 1,111$) show considerably wider spread, which reflects the well-documented molecular heterogeneity of breast cancer across patients. Notably, there is a small number of normal samples that drift toward the tumor cluster, which matches a pattern previously observed in TCGA where adjacent normal tissue can harbor tumor-like expression profiles (Huang et al., 2016). Overall, the PCA confirms that the dataset is high quality and that tumor vs. normal status is the primary driver of global gene expression differences, justifying the downstream differential expression analysis.

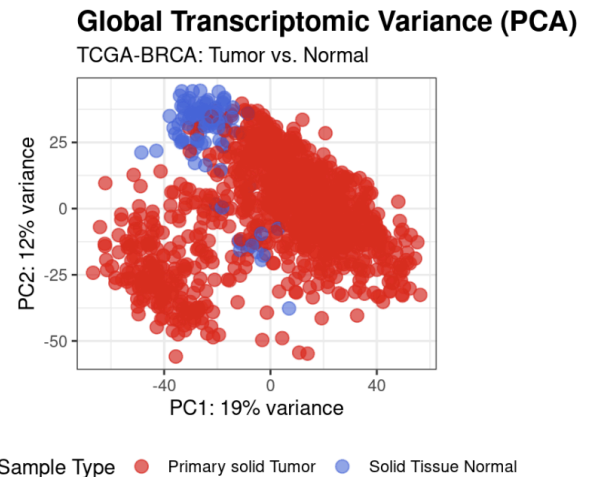


Figure 1: PCA of VST-normalized expression data across 1,224 samples.

2. DESeq2 Identifies Widespread Transcriptomic Dysregulation

Applying the dual threshold of adjusted $p\text{-value} < 0.001$ and $|\log_2\text{FC}| > 2$ to the DESeq2 results identified a large number of statistically and biologically significant DEGs across the 56,709 genes tested. The volcano plot in Figure 2 visualizes the full distribution of results, with each point representing one gene plotted by its $\log_2$ fold change on the x-axis and its $-\log_{10}$ adjusted p-value on the y-axis. Red points in the upper left and upper right quadrants represent top hits, or genes that passed both thresholds simultaneously.

Among the most significant upregulated genes in primary tumor relative to normal tissue is MUC1, a well-characterized oncogene that showed a $\log_2$ fold change of approximately +7 and a $-\log_{10}$ adjusted p-value exceeding 300, placing it among the most extreme findings in the dataset. PIP, a tissue marker

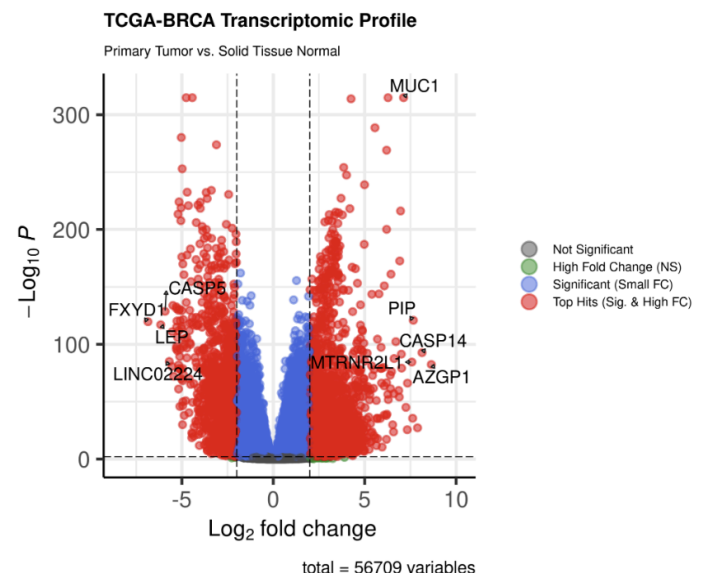


Figure 2: Volcano plot of differential expression results across 56,709 genes. Red dots represent top hits ($p\text{-value} < 0.01$ and $|\log_2\text{FC}| > 2$).

previously associated with tumor proliferation in breast cancer, was also among the top upregulated hits. On the downregulated side, FXYD1, LEP, and CASP5, which are genes associated with normal adipose and epithelial tissue function, showed substantial suppression in tumor tissue, consistent with the cancer identity. In summary, the large number of DEGs in both directions reflects the widespread gene expression changes that define breast cancer.

3. Top 50 DEGs Cleanly Separate Tumor from Normal

To visualize the collective expression signature of the most significant DEGs, a clustered heatmap was generated using the top 50 genes ranked by adjusted p-value, displayed across a representative subsample of 200 patients (100 tumor, 100 normal). As shown in Figure 3, unsupervised hierarchical clustering, applied independently to both rows (genes) and columns (samples) without any clinical label information, cleanly separated the 200 samples into two groups that align precisely with tumor and normal tissue type. This result demonstrates that the expression signal captured by the top 50 DEGs alone is strong enough for an unsupervised algorithm to recover the biological groupings without any prior knowledge of sample identity.

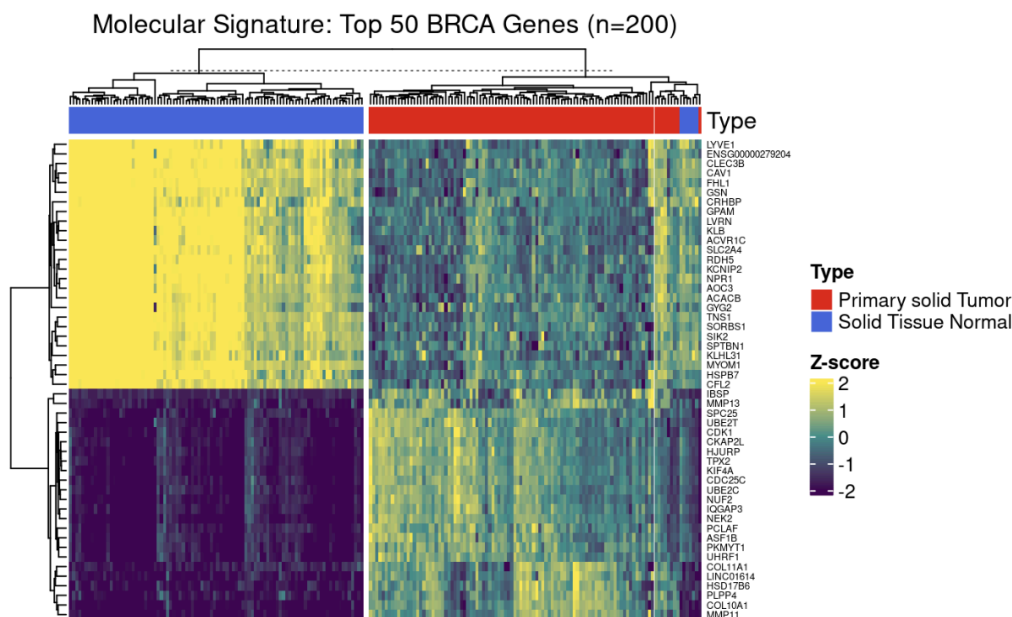


Figure 3: Hierarchical clustering heatmap of the top 50 DEGs across 200 patient subsample.

Two distinct gene clusters are visible along the row axis. The upper cluster, which includes genes such as LYVE1, CAV1, and GSN, shows high expression (yellow) in normal tissue and suppressed expression (violet) in tumor samples, suggesting that these genes are characteristic of healthy breast tissue and are lost during the transition to a cancerous state. The lower cluster, which includes cell cycle and matrix remodeling genes such as CDK1, UBE2C, and MMP11, shows the inverse pattern (suppressed in normal tissue and highly expressed in tumor), which is consistent with the proliferative and invasive phenotype of cancer cells (Cancer Genome Atlas Network, 2012).

4. MMP11 Expression Varies Significantly Across PAM50 Subtypes

To explore whether the expression of a top DEG varied across established molecular subtypes, MMP11 was selected for a clinical deep-dive, given its biological relevance as a matrix remodeling

enzyme (Sathyanarayanan et al., 2020) and its prominent position in the heatmap. VST-normalized MMP11 expression was compared across the four PAM50 subtypes (Basal, Her2-enriched, Luminal A, and Luminal B) across all 1,099 tumor samples that had subtype annotation available. As shown in Figure 4, MMP11 expression varies meaningfully across subtypes. The Her2-enriched subtype shows the highest median expression, followed closely by Luminal A and Luminal B. Meanwhile, the Basal subtype shows a notably lower median and a tighter distribution. This pattern is clinically interesting because Basal-like breast cancer is generally considered the most aggressive subtype with the poorest prognosis, yet it doesn't show the highest MMP11 expression. This suggests that MMP11's role in tumor biology is subtype-dependent rather than a universal marker of aggressiveness. The broad distribution of expression values within each subtype further indicates that individual patient variability in MMP11 activity is substantial even within molecularly defined groups.

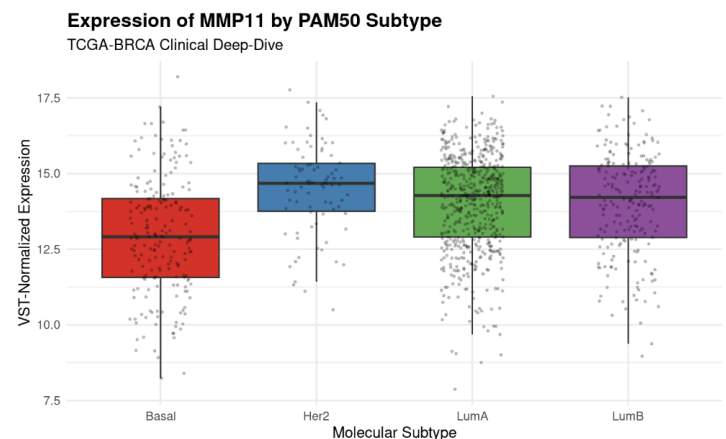


Figure 4: VST-normalized MMP11 expression stratified by PAM50 molecular subtype across 1,099 tumor samples.

Conclusion

This analysis successfully identified thousands of statistically and biologically significant differentially expressed genes between primary breast tumor and healthy adjacent tissue in the TCGA-BRCA cohort, demonstrating that tumor vs. normal status is the dominant driver of transcriptomic variation across this dataset. The convergence of evidence across all four figures collectively supports the robustness of the DESeq2 pipeline applied here. MMP11 emerged as a candidate biomarker of interest, showing broad upregulation in tumor tissue overall. However, its expression differed meaningfully across PAM50 subtypes, suggesting that its clinical utility may be subtype-specific rather than universal.

There were also several limitations of this analysis. The most notable is the imbalance between tumor ($n = 1,111$) and normal ($n = 113$) samples, which is a structural feature of the TCGA-BRCA cohort rather than an analytical choice. While DESeq2 is robust to unequal group sizes, this imbalance may affect sensitivity and limit the ability to perform a fully paired analysis controlling for individual patient effects. Additionally, while the dual threshold of adjusted $p\text{-value} < 0.01$ and $|\log_2\text{FC}| > 2$ is stringent by conventional standards, the scale of the dataset means that thousands of genes still pass these criteria, and not all will be biologically meaningful in a clinical context. Candidate genes identified here would require experimental validation before any biomarker claims could be made.

For future directions, a natural extension of this work would be a Kaplan-Meier survival analysis to test whether MMP11 expression level is associated with overall patient survival. This finding would provide stronger evidence for MMP11 as a clinically relevant biomarker. Additionally, TCGAbiolinks could be used to retrieve matched DNA methylation or copy number variation data from the same cohort, allowing for a multi-omic analysis in R to determine whether the transcriptomic differences identified here are driven by upstream epigenetic or genomic alterations. This could provide a more complete picture of the molecular drivers of breast cancer.

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